

Knowledge:

1. Divide $6x^3 - x^2 + 5$ by $x - 3$ using long division. Write the corresponding division statement.

21

Total
123

$$\begin{array}{r} x-3 \overline{) 6x^3 - x^2 + 0x + 5} \\ \underline{6x^3 + 17x + 51} \\ - \rightarrow 6x^3 - 18x^2 \end{array}$$

(3)

$$\underline{17x^2 + 0x}$$

$$\begin{array}{r} - \rightarrow 17x^2 - 51x \\ \underline{51x + 5} \\ - \rightarrow 51x - 153 \end{array}$$

2. Using the Remainder Theorem, determine the remainder when $3x^3 + 2x^2 - 3x - 50$ is divided by $2x - 5$.

$$\begin{aligned} 2x - 5 &= 0 \\ x &= \frac{5}{2} \end{aligned}$$

(2)

$$f\left(\frac{5}{2}\right) = 3\left(\frac{5}{2}\right)^3 + 2\left(\frac{5}{2}\right)^2 - 3\left(\frac{5}{2}\right) - 50$$

$$= 1.875$$

$$= \boxed{\frac{15}{8}}$$

3. Factor.

a) $4x^2 - 17x + 15$

$$= (4x - 5)(x - 3)$$

(2)

b) $9x^4 + 32x^2 - 16$

$$= (9x^2 - 4)(x^2 + 4)$$

$$\begin{array}{r} 9x^2 \quad -4 \\ x^2 \quad \quad 4 \end{array}$$

$$= (3x - 2)(3x + 2)(x^2 + 4)$$

4 - 5

1 - 3

c) $27x^3 + 216$

$$(3x)^3 + 6^3$$

(2)

$$(3x + 6)(9x^2 + 18x + 36)$$

$$27(x + 2)(x^2 + 2x + 4)$$

$$\rightarrow = (8x - 15y)(x + 6y)$$

d) $81(x - y)^2 - 49(3y - x)^2$

$$(9(x - y))^2 - (7(3y - x))^2$$

$$(9(x - y) - 7(3y - x))(9(x - y) + 7(3y - x))$$

$$= (9x - 9y - 21y + 7x)(9x - 9y + 21y - 7x)$$

$$= (16x - 30y)(2x + 12y)$$

e) $x^4 - 5x^3 + 2x^2 + 20x - 24 = 0$

$$\begin{array}{r} 3 \overline{) 1 - 5 2 20 - 24} \\ \underline{3 - 6 - 12 24} \\ -2 1 - 2 - 4 8 0 \\ \underline{-2 2 8 - 8} \\ 1 - 4 4 0 \end{array}$$

(3)

$$(x-3)(x+2)(x-2)$$

$$(x-3)(x+2)(x^2-4x+4)$$

Application:

1. Determine the height of a cylindrical can that has a volume of $(8\pi x^3 - 20\pi x^2 + 14\pi x - 3\pi) \text{ cm}^3$.
($V = \pi r^2 h$)

$$\pi (8x^3 - 20x^2 + 14x - 3)$$

$$\begin{array}{r} (3) \frac{3}{2} \overline{) 8 - 20 14 - 3} \\ \underline{12 - 12 3} \\ 8 - 8 2 0 \end{array}$$

$$\pi (2x-3)^2$$

$$h = 2x - 3$$

$$(x + \frac{3}{2})(8x^2 - 8x + 2)$$

$$(2x-3)(4x^2-4x+1)$$

2. When $2x^3 - mx^2 + nx - 78$ is divided by $x-1$, the remainder is -84 , and $x-3$ is a factor. Determine the values of m and n .

$$-2 - m - n - 78 = -84 \rightarrow -m - n = -4$$

$$2(27) = m(9) + 3n - 78 = 0 \rightarrow 9m + 3n = 78$$

$$3m + n = 26$$

$$-4m = -46$$

$$m = -1$$

$$m + n = -4$$

$$-1 + n = -4$$

$$n = -3$$

$$\begin{array}{l} m = -1 \\ n = -3 \end{array}$$